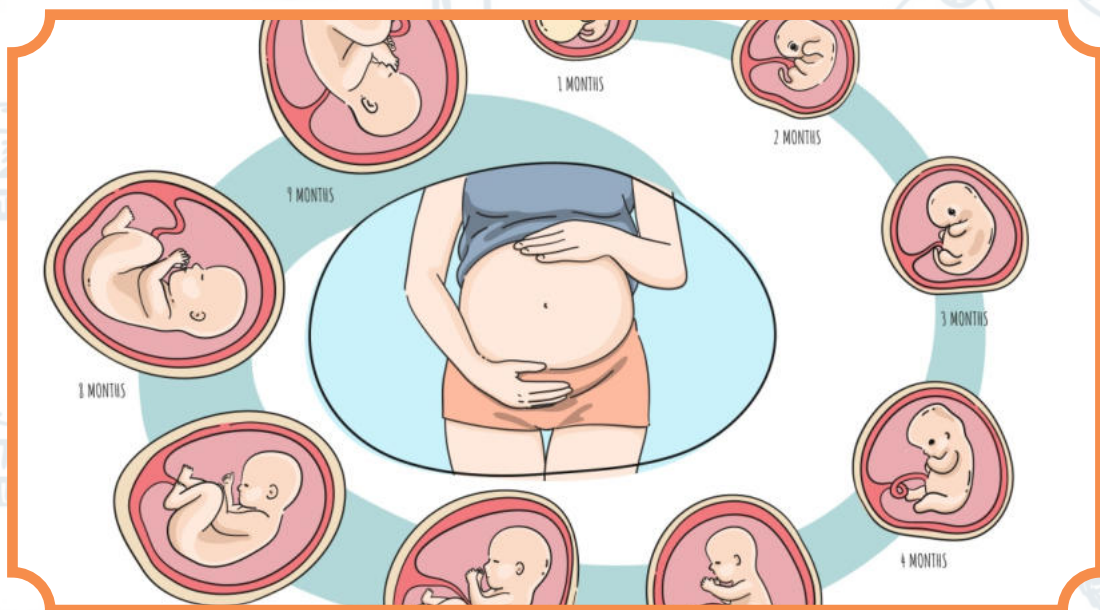


UNIT

20

REPRODUCTION



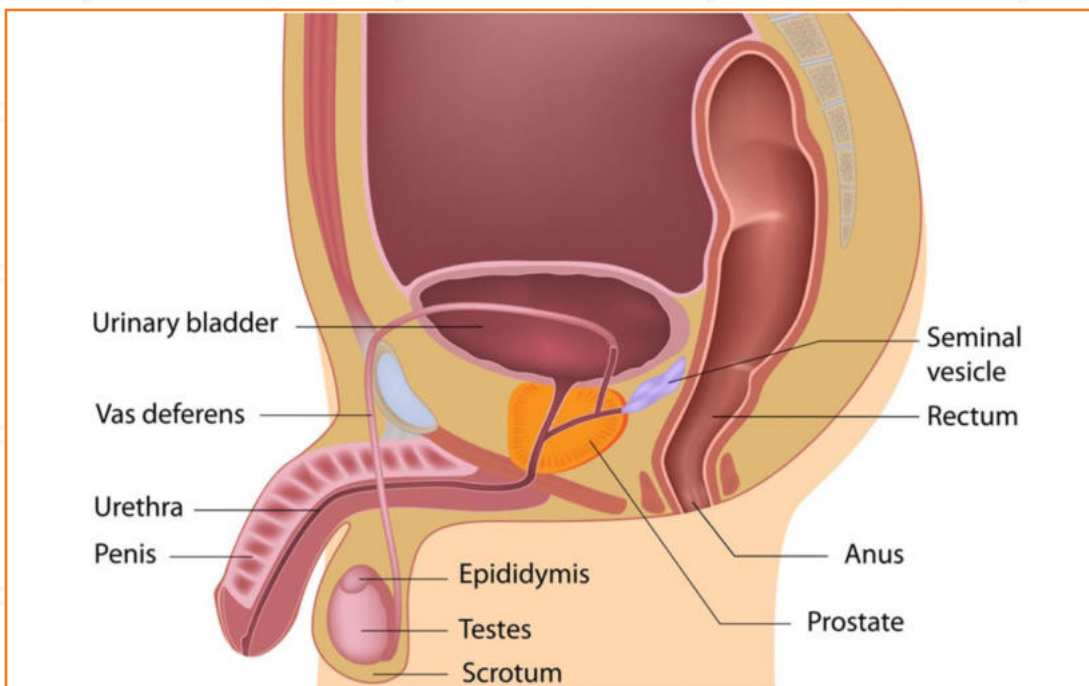


Human Reproductive System

- **General Overview**
- The human reproductive system enables reproduction.
- Humans are **unisexual** (dioecious/heterogametic), meaning individuals are either **male** or **female**.
- Males have **testes**, and females have **ovaries**.

Male Reproductive System

- **Function:** Produces and delivers **sperm** to the female reproductive system.
- **Primary organs:**
 - **Testes** produce sperm and the hormone **testosterone**.
- **Accessory structures:**
 - **Epididymis**—stores and matures sperm.
 - **Vas deferens**—transports sperm.
 - **Seminal vesicles, prostate gland, and bulbourethral glands** add fluids to form **semen**.
 - **Penis**—organ for delivering sperm into the female reproductive tract.



Female Reproductive System

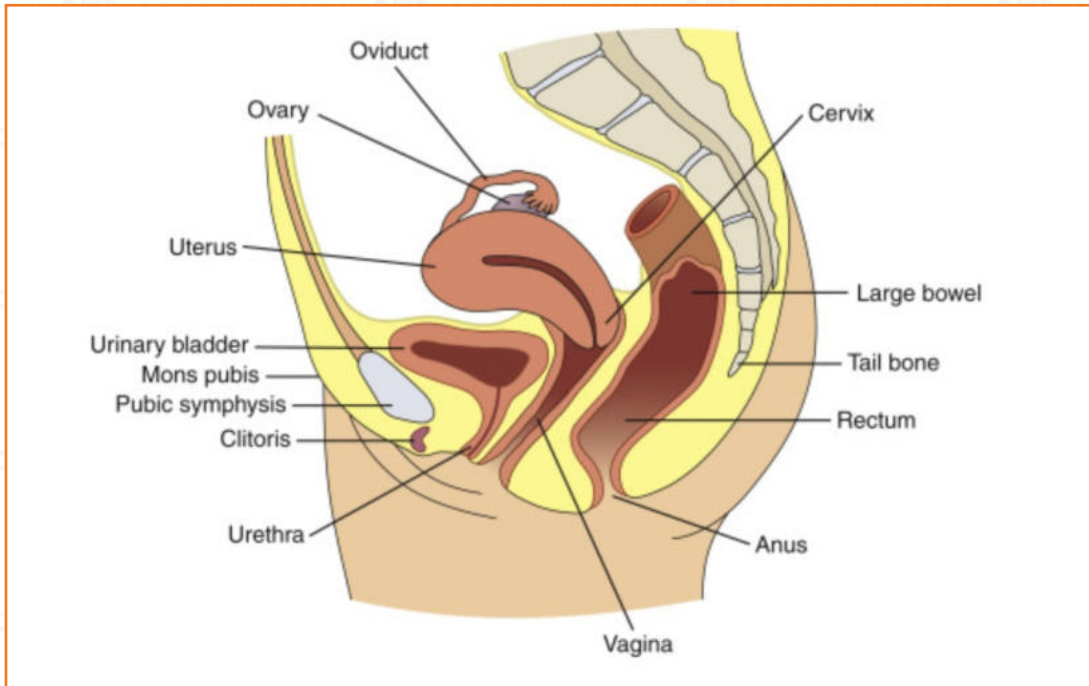
- **Function:** Produces **eggs (ova)**, facilitates **fertilization**, and supports **fetal development**.
- **Primary organs:**
 - **Ovaries** produce ova and the hormones **estrogen** and **progesterone**.





Accessory structures:

- **Fallopian tubes**—site of fertilization, transport egg to uterus.
- **Uterus**—supports embryo/fetus during pregnancy.
- **Vagina**—receives sperm, birth canal during delivery.
- **Cervix**—connects uterus to vagina.



Key Functions

- **Male:** Sperm production and delivery.
- **Female:** Egg production, fertilization, and pregnancy support.

Male Reproductive System

1. Overview

- Composed of **external** and **internal genitalia**.
- **Primary function:** Production, maturation, storage, and delivery of **sperm**, and secretion of **male sex hormones**.

2. External Genitalia

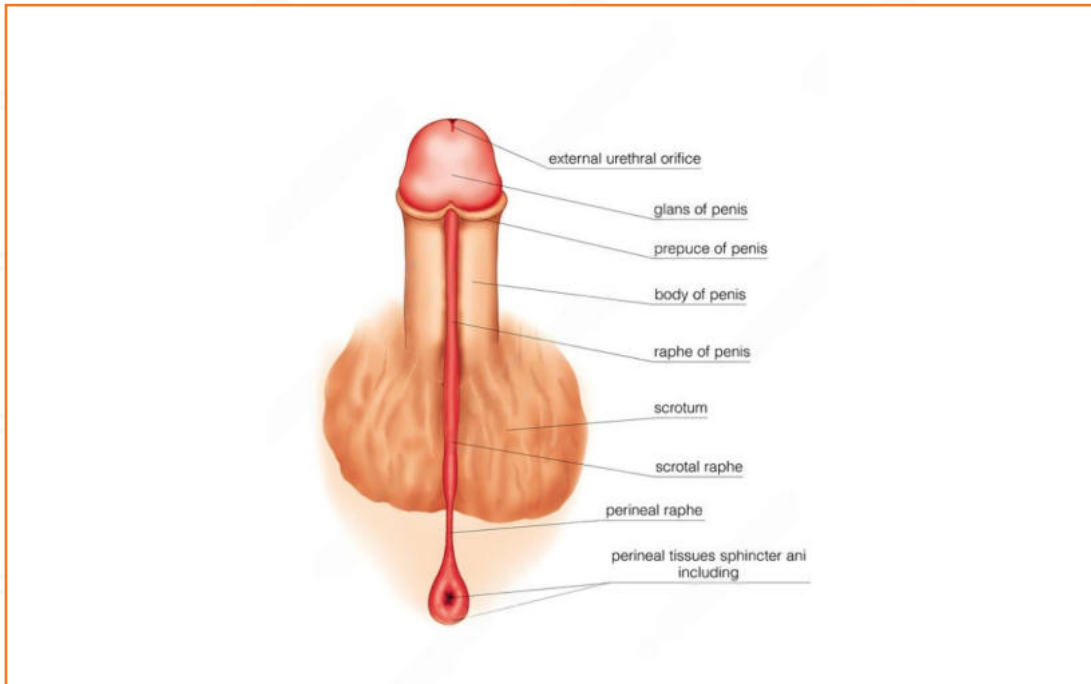
- **Penis:**
 - Copulatory organ.
 - Parts:
 - **Glans** (tip): Contains urethral opening, very sensitive.
 - **Shaft:** Contains erectile tissue.
 - **Root:** Anchors penis to the pelvic region.
 - **Foreskin** may cover the glans.





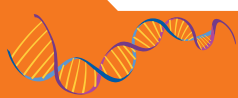
Scrotum:

- Pouch of skin that houses the **testes**.
- Maintains temperature around **34°C**, ideal for **spermatogenesis**.



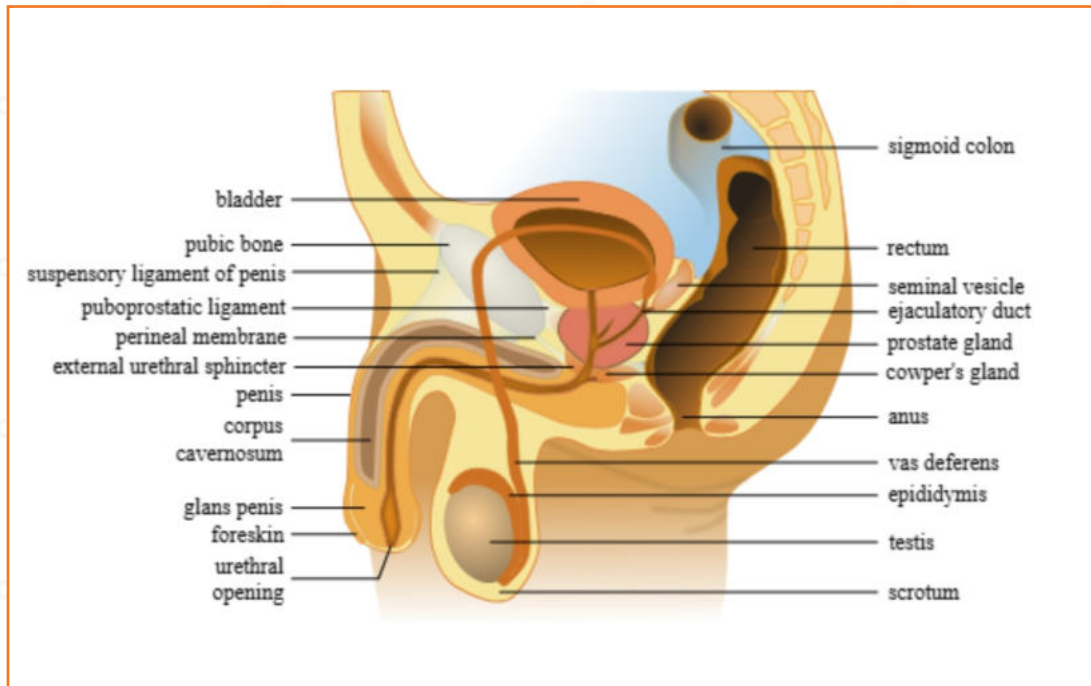
3. Internal Genitalia

- **Gonads (Testes)**
- Develop in abdomen; descend into **scrotum** before birth.
- Functions:
 - **Sperm production** (in **seminiferous tubules**).
 - Secretion of **testosterone**.
- **Sertoli cells**: Support and nourish developing sperm.
- **Spermatogenesis**: Process of sperm formation.
- **Accessory Ducts**
- **Rete testis**: Network of tubules collecting sperm from seminiferous tubules.
- **Efferent ducts**: Connect rete testis to epididymis.
- **Epididymis**:
 - Site of **sperm maturation and storage**.
 - Sperm becomes motile here.
- **Vas deferens (Ductus deferens)**:
 - Thick tube conducting sperm from epididymis to **ejaculatory duct**.
 - Has an expanded region called **ampulla** (reservoir).





- **Ejaculatory duct**
 - Formed by fusion of **seminal vesicle duct** and **ampulla**.
 - Transfers semen into **urethra**.
- **Urethra**
 - Common passage for **urine and semen**.
 - Known as the **urinogenital duct**.



4. Accessory Glands

- **Seminal Vesicles:**
 - Contribute ~**60%** of semen volume.
 - Secrete fructose-rich fluid for sperm energy.
- **Prostate Gland:**
 - Largest semen-producing gland.
 - Secretes alkaline fluid into urethra.
- **Bulbourethral (Cowper's) Glands:**
 - Secrete **clear mucus** before ejaculation.
 - Neutralizes acidic urine in urethra.

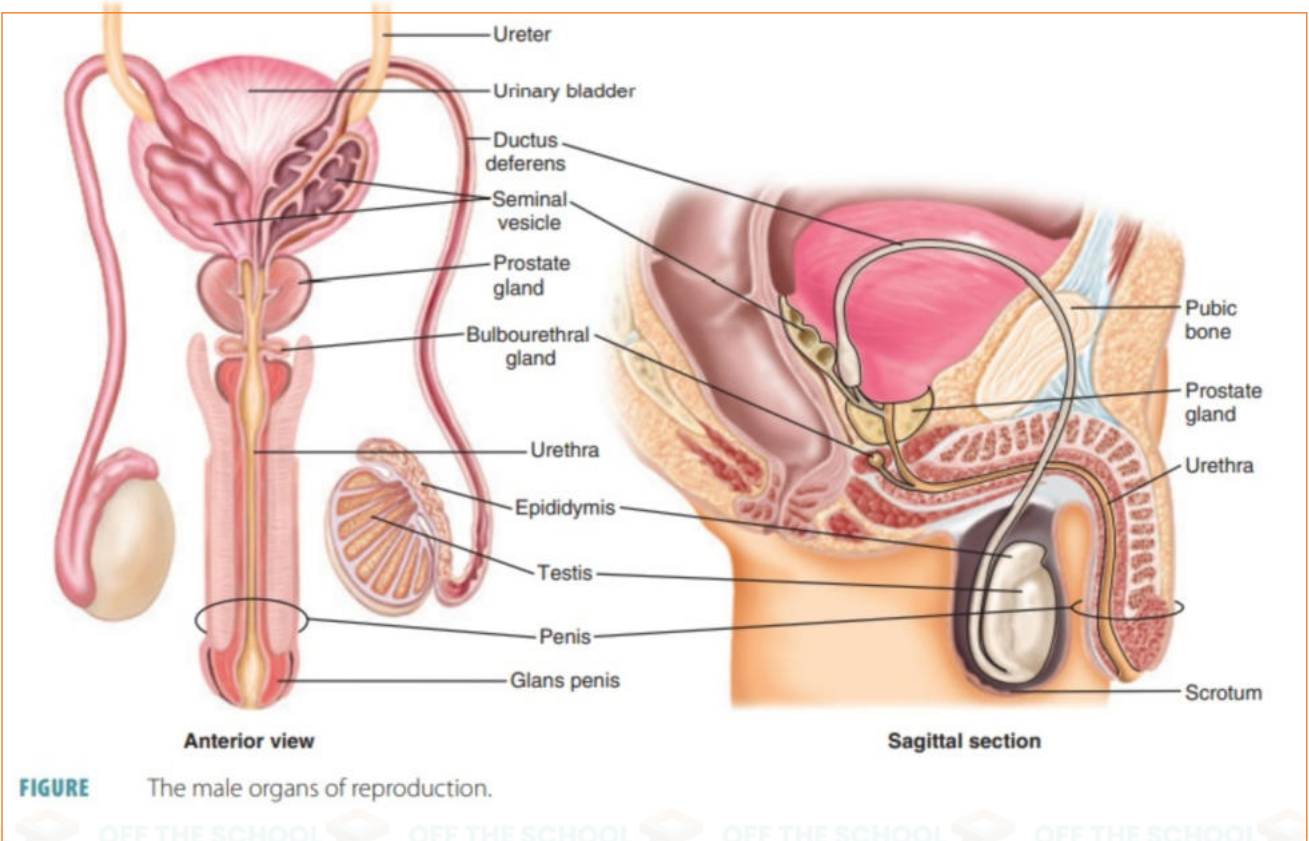
5. Semen Composition

- Volume per ejaculation: **2 - 5 ml**.
- Sperm count: **200 - 300 million sperm** per ejaculate.





ORGAN	FUNCTION
Testes	Produce sperm and male sex hormones
Epididymis	Maturation and temporary storage of sperm
Vas deferens	Transport and store sperm
Seminal Vesicles	Add nutrient-rich fluid to semen
Prostate Gland	Secretes alkaline fluid to activate sperm
Urethra	Conducts sperm and urine
Bulbourethral Glands	Secrete mucus to neutralize urethra



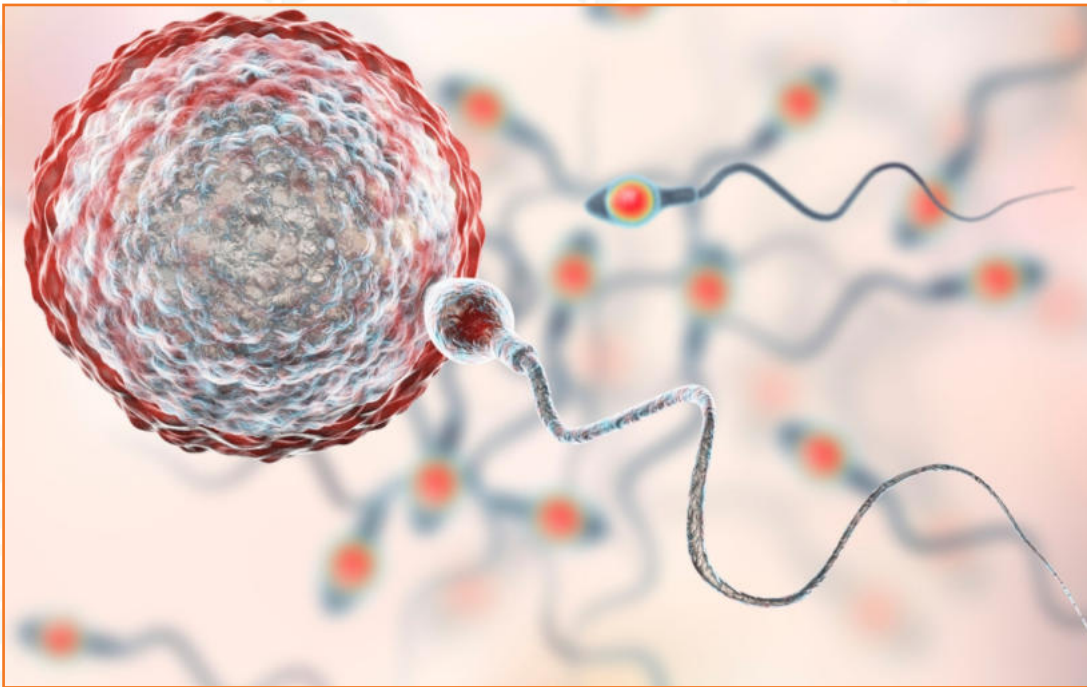


Male Reproductive System & Physiology: Spermatogenesis

- **Definition**
- **Spermatogenesis:** The process of **sperm production** occurring in the **seminiferous tubules** of the **testes**.

Location

- **Seminiferous tubules** within the testes.
- Involves specialized cells called **Sertoli cells** that nourish developing sperm.



Phases of Spermatogenesis

1. Proliferation Phase (Mitosis)

- Begins with **spermatogonia (2n)**: Undifferentiated germ cells.
- Spermatogonia undergo **mitosis** to:
 - Maintain a pool of germ cells.
 - Produce **primary spermatocytes (2n)**.

2. Growth Phase

- **Primary spermatocytes** replicate their DNA in preparation for meiosis.
- Each now contains **double the DNA**.

3. Meiosis I

- **Primary spermatocytes (2n)** divide to form **secondary spermatocytes (n)**.
- Chromosome number is **halved**.
- Introduces **genetic variation**.





4. Meiosis II

- **Secondary spermatocytes (n)** divide to form **spermatids (n)**.
- Four **haploid spermatids** are produced from each primary spermatocyte.

5. Spermiogenesis

- Transformation of **spermatids** into **spermatozoa (mature sperm cells)**.
- Key changes:
 - **Nucleus condenses**.
 - **Acrosome forms** (contains enzymes to penetrate egg).
 - **Flagellum develops** (for motility).
 - Cytoplasm is reduced.

Final Stage

- Mature **spermatozoa** are released into the **lumen** of seminiferous tubules (process called **spermiation**).
- Travel to the **epididymis** where they:
 - Gain **motility**.
 - Achieve **full maturity** for fertilization.

STAGE	CELL TYPE	CHROMOSOME NUMBER	PROCESS
Mitosis	Spermatogonia / Primary Spermatocyte	Diploid (2n)	Cell division and renewal
DNA Replication	Primary Spermatocyte	Diploid (2n, replicated)	Prepares for meiosis
Meiosis I	Primary Secondary Spermatocyte	Haploid (n)	Chromosome number halved
Meiosis II	Secondary Spermatids	Haploid (n)	4 non-motile haploid spermatids
Spermiogenesis	Spermatids Spermatozoa	Haploid (n)	Morphological maturation





Hormonal Regulation in Males

The male reproductive system is regulated by a network of hormones that control **sperm production, testosterone secretion**, and the development of **secondary sexual characteristics**.

HORMONE	PRODUCED BY	FUNCTION
Testosterone	Leydig cells (in testes)	-Primary male sex hormone - Development of male reproductive organs (testes, prostate, seminal vesicles) - Promotes secondary sexual characteristics (e.g., deep voice, facial/body hair, muscle mass) - Stimulates spermatogenesis
Follicle-Stimulating Hormone (FSH)	Anterior Pituitary	- Acts on Sertoli cells in seminiferous tubules - Promotes maturation of sperm cells - Stimulates production of sperm-supporting proteins
Luteinizing Hormone (LH)	Anterior Pituitary	- Stimulates Leydig cells to produce testosterone
Inhibin	Sertoli cells (in testes)	- Inhibits FSH secretion to regulate sperm production
Gonadotropin-Releasing Hormone (GnRH)	Hypothalamus	- Stimulates anterior pituitary to release FSH and LH - Controlled by a negative feedback loop from testosterone

Hormonal Feedback Loop

- GnRH** from the hypothalamus stimulates the release of:
 - **FSH** → acts on Sertoli cells → supports **spermatogenesis**
 - **LH** → stimulates Leydig cells → produces **testosterone**
- Testosterone**:
 - Promotes male traits and sperm production.
 - **Inhibits GnRH, FSH, and LH** when levels are high (**negative feedback**).





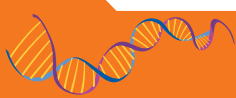
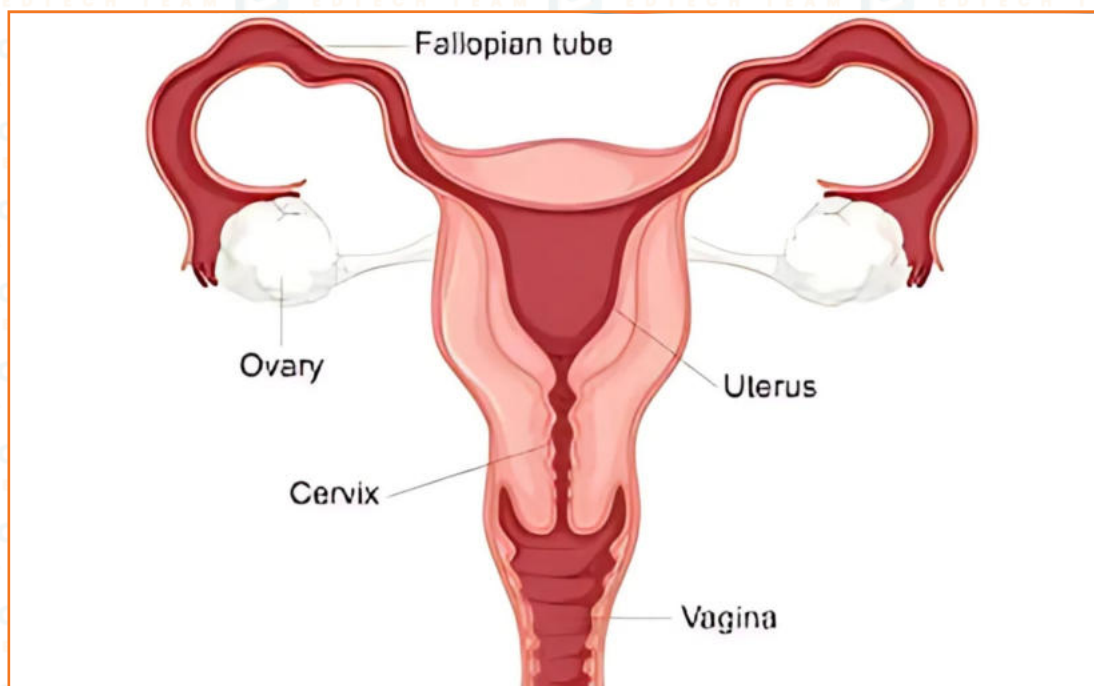
3. Inhibin:

- Released from Sertoli cells.
- Specifically inhibits **FSH** to fine-tune sperm production.

HORMONE	TARGET	MAIN EFFECT
GnRH	Pituitary gland	Stimulates FSH and LH release
FSH	Sertoli cells	Initiates spermatogenesis
LH	Leydig cells	Stimulates testosterone production
Testosterone	Male body tissues	Develops male traits and supports sperm production
Inhibin	Pituitary gland	Inhibits FSH secretion

Female reproductive system

The female reproductive system is composed of **four main parts** that work together for the production of egg cells, fertilization, and fetal development.

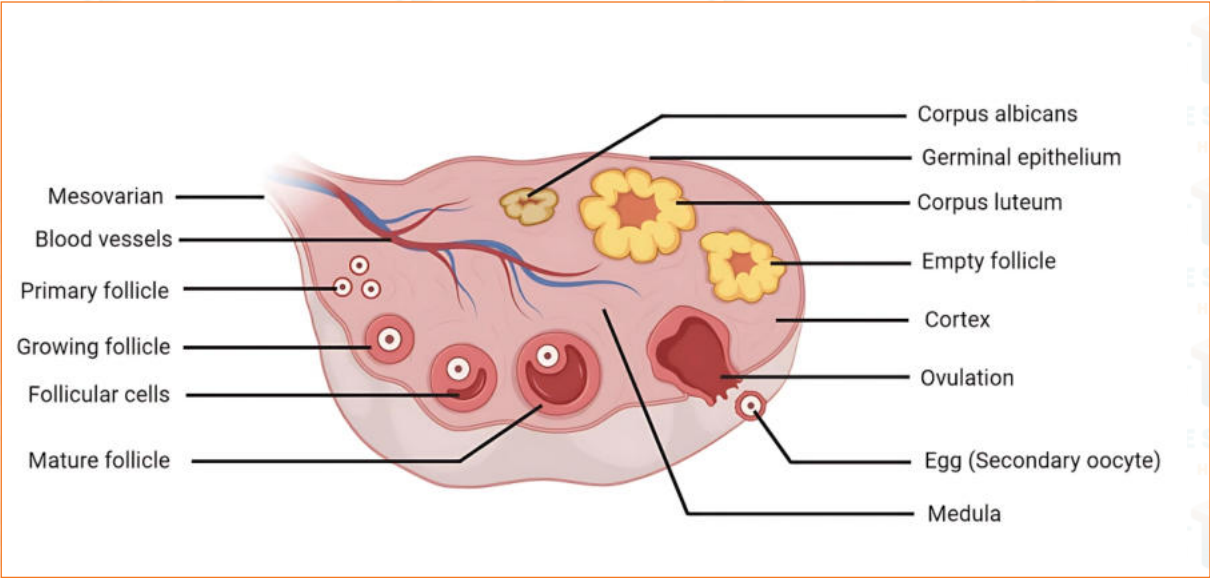


Main Components and Their Functions

ORGAN	FUNCTION
Ovaries	Produce eggs (ova) and female sex hormones (e.g., estrogen)
Oviducts	Conduct eggs from ovary to uterus; site of fertilization
Uterus	Supports embryo development and fetal growth
Vagina	Receives sperm during intercourse; acts as birth canal

Ovaries (Female Gonads)

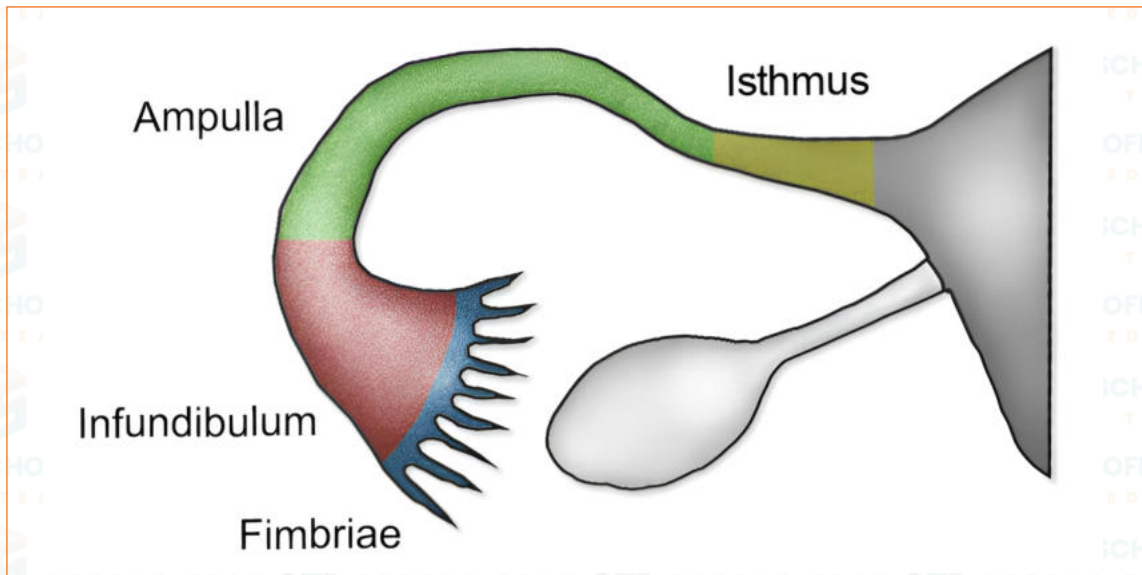
- Located in the **abdominal cavity**, attached near the uterus by a **mesentery**.
- Enclosed in a **protective capsule**.
- Contains **follicles**: each follicle has one **egg (oocyte)** surrounded by **supporting follicle cells**.
- Around **200,000 follicles** present at birth, only **a few hundred** mature and release eggs in a women’s lifetime.
- Follicle cells also produce **estrogen**, a key hormone in the menstrual cycle and secondary sexual characteristics.





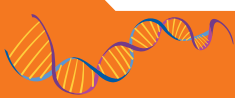
Oviducts (Fallopian Tubes / Uterine Tubes)

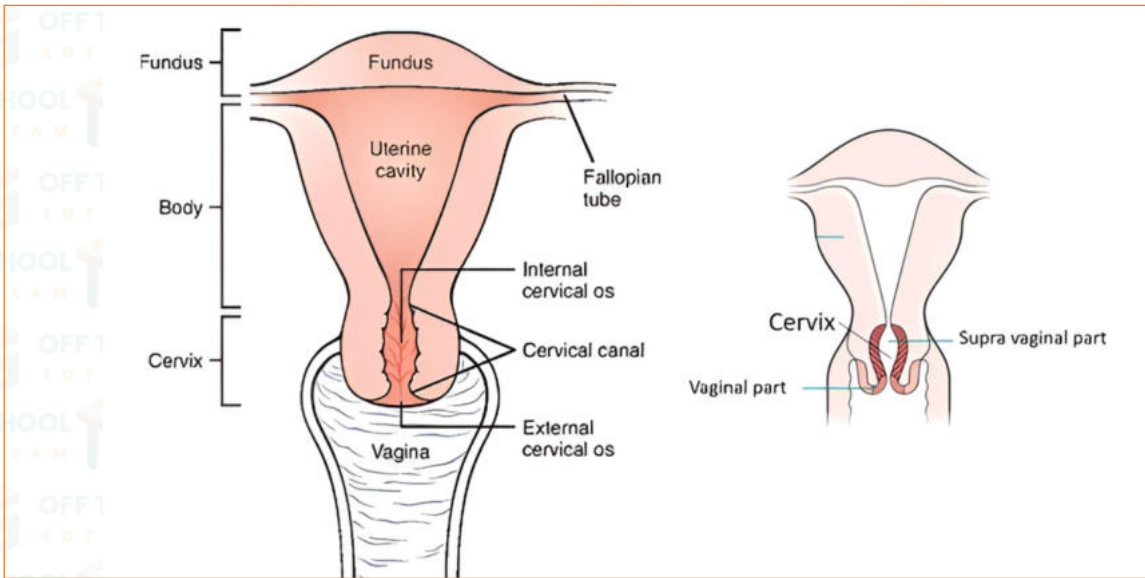
- Tubes ~10 cm long that transport the egg to the uterus.
- Ends are fringed with **fimbriae** (ciliated finger-like structures) near the ovaries.
- The **inner lining has cilia** that help move the egg.
- **Fertilization usually occurs** here.



Uterus (Womb)

- Muscular, inverted **pear-shaped organ**.
- Located **between bladder and rectum**.
- Divided into 3 regions:
 - **Fundus**: Upper rounded part
 - **Body**: Main central portion
 - **Cervix**: Lower narrow passage leading to the vagina
- Uterine Wall Layers:
 - **Endometrium**: Inner epithelial lining, rich in blood vessels, supports embryo
 - **Myometrium**: Thick middle **muscular** layer
 - **Perimetrium**: Outer thin protective covering
- **Cervix**: Narrow passage between uterus and vagina; secretes mucus and dilates during childbirth.





Vagina

- A muscular, elastic **canal**.
- Functions:
 - **Receives sperm** during intercourse.
 - **Serves as the birth canal** during delivery of the fetus.

Quick Reference Table

ORGAN	FUNCTION
Ovaries	Produce eggs and hormones (e.g., estrogen)
Oviducts	Conduct egg; site of fertilization
Uterus	Houses and nourishes the embryo/fetus
Vagina	Sperm reception; birth canal

The Ovarian Cycle

- Human females **do not** have an oestrous cycle like lower mammals.
- Instead, **one egg is released approximately every 28 days** this is the **ovarian cycle**.



Stages of Follicular Development

1. Primary Follicle:

- Contains a **primary oocyte (2n)** surrounded by epithelial cells.

2. Secondary Follicle:

- Characterized by the appearance of **follicular fluid pools** around the oocyte.

3. Vesicular (Graafian) Follicle:

- Large, mature follicle with a **fluid-filled cavity** that bulges at the ovarian surface.

Hormonal Regulation:

- Controlled by:
 - FSH (Follicle-Stimulating Hormone)** stimulates follicle growth.
 - LH (Luteinizing Hormone)** triggers ovulation and corpus luteum formation.
- Both are secreted by the **anterior pituitary**.

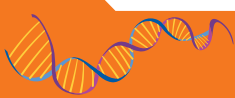
Oogenesis (formation of Egg)

Process Overview

- Germ cells** divide mitotically → form **oogonia (2n)**.
- Oogonia develop into **primary oocytes (2n)**.
- Meiosis I:**
 - Produces a **secondary oocyte (n)** and a **first polar body (n)** (unequal division).
- During **ovulation**, the **secondary oocyte is released**.
- Fertilization** (if it occurs):
 - Triggers **completion of meiosis II**.
 - Forms a **mature egg (n)** and a **second polar body (n)**.
- Zygote formation:**
 - Fusion of sperm (n) and egg (n) → **zygote (2n)** with **46 chromosomes**.

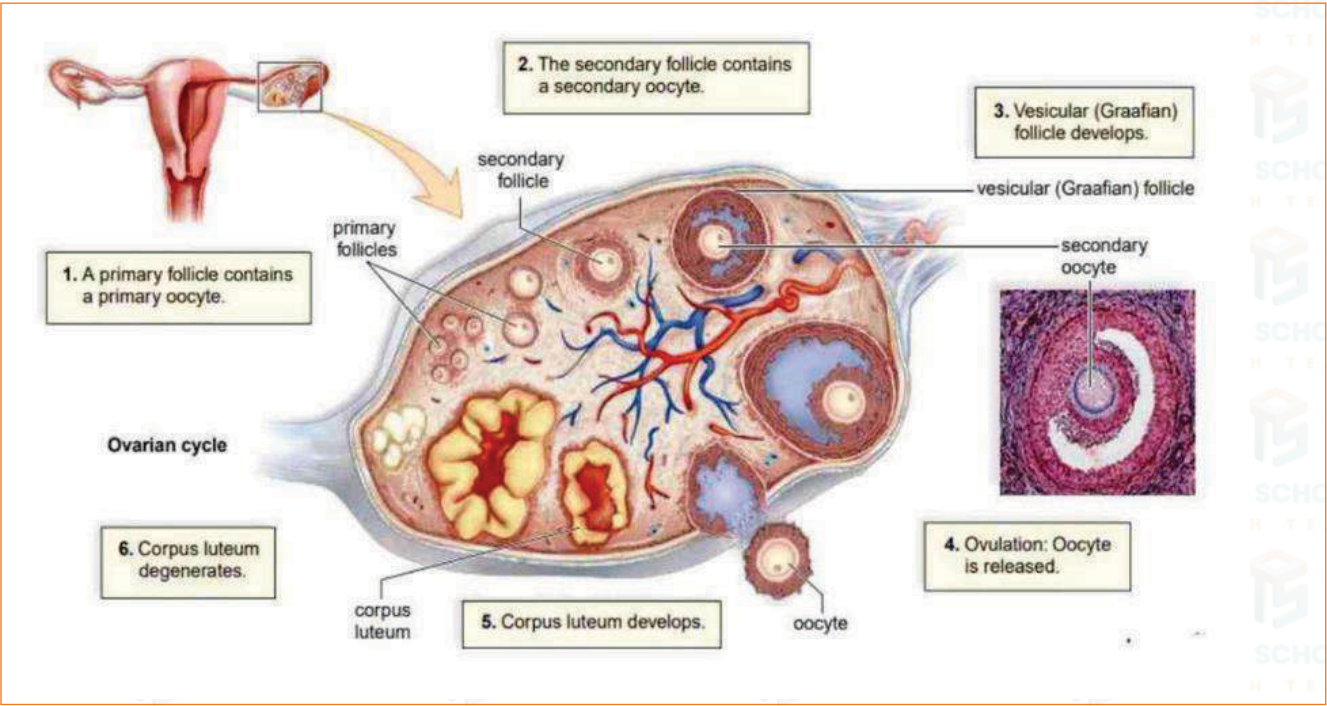
Corpus Luteum :

- After ovulation, the ruptured Graafian follicle transforms into the **corpus luteum**.
- Secretes **progesterone and estrogen**.
- Degenerates if pregnancy does not occur.



PHASE	DAYS	HARMONES	KEY EVENTS
Menstrual	1 — 5	Low Estrogen & Progesterone	Endometrium sheds menstruation occurs
Proliferative	6 — 13	↑ Estrogen	Endometrium rebuilds ; follicles mature
Ovulation	~14	LH Surge	Egg released from Graafian follicle
Secretory	15 — 28	↑ Progesterone	Endometrium thickens and matures potential implantation

- If **no fertilization**, hormone levels drop ↑ **endometrium breaks down**, starting a new cycle.





1. Menstruation Phase (Days 1–5)

- Begins with **menstrual bleeding**.
- Occurs due to **decline in progesterone** as the **corpus luteum degenerates**.
- The **endometrium (uterine lining)** is shed.
- **First day of bleeding = Day 1 of new cycle.**

2. Proliferative / Pre-ovulatory Phase (Days 6 – 14)

- Rising **FSH stimulates growth of ovarian follicles**.
- A **dominant follicle** matures into a **Graafian follicle**.
- Follicle secretes **estrogen**, which:
 - **Thickens and vascularizes the endometrium.**
 - Makes **cervical mucus thin and sperm-friendly**.
 - Causes **negative feedback** on FSH and triggers **LH surge**.

Ovulation (~Day 14):

- Sudden **LH surge** ↑ Graafian follicle ruptures ↑ **Secondary oocyte released**.
- Ruptured follicle becomes **corpus luteum**.

3. Secretory / Post-ovulatory Phase (Days 15–28)

- **Corpus luteum** secretes **progesterone**.
- **Endometrium becomes thick, glandular, and nutrient-rich** for potential implantation.
- If **fertilization does not occur**:
 - Corpus luteum degenerates → ↓ Progesterone.
 - Triggers **endometrial shedding** → **menstruation** begins again.

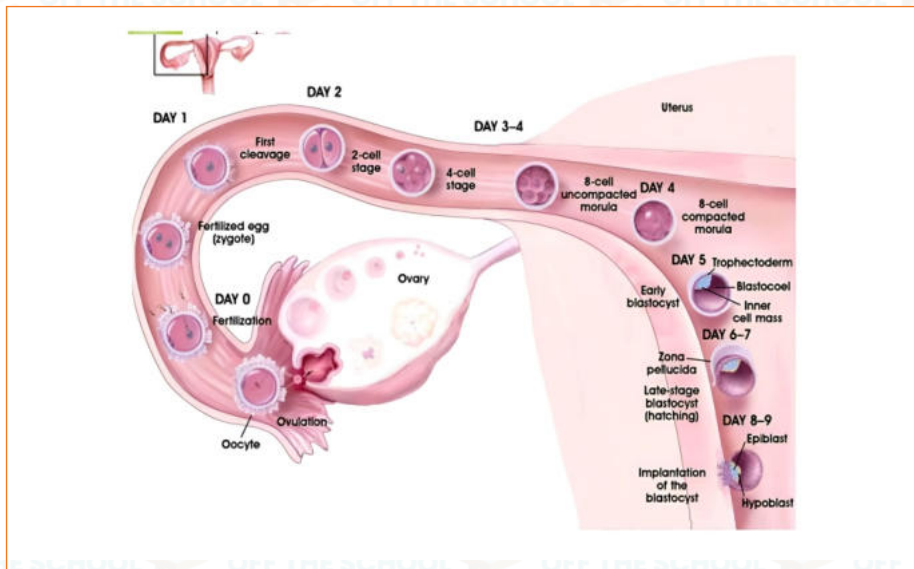
HORMONE	SOURCE	FUNCTION
FSH	Anterior pituitary	Stimulates follicle growth
LH	Anterior pituitary	Triggers ovulation and corpus luteum formation
Estrogen	Follicular cells	Rebuilds endometrium, suppresses FSH, triggers LH surge
Progesterone	Corpus luteum	Maintains thick endometrium for implantation





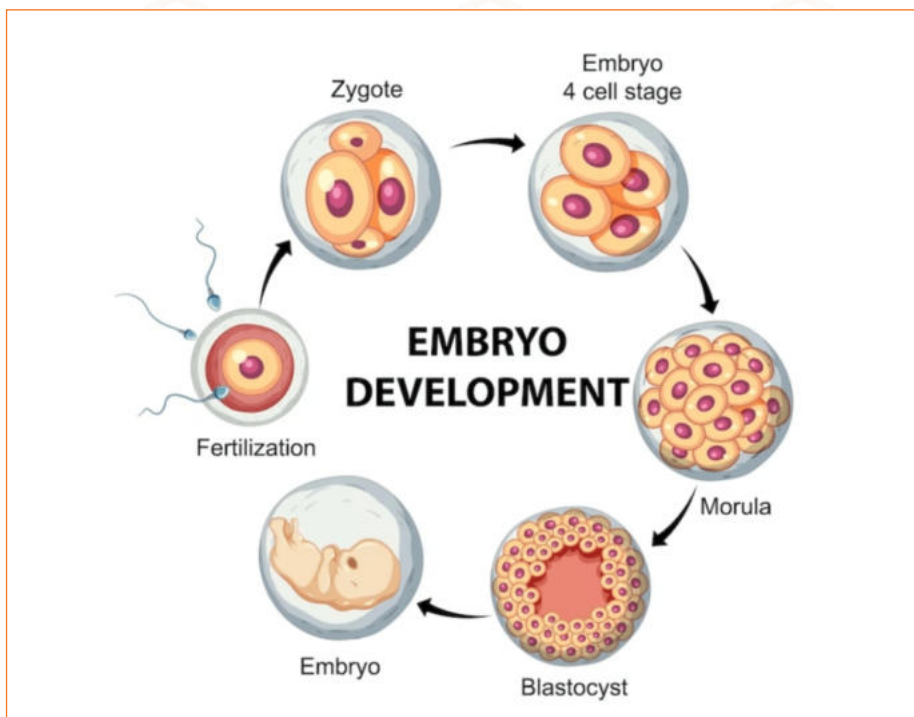
Fertilization

- Occurs when a **sperm cell fuses with a secondary oocyte** (egg) in the **oviduct (fallopian tube)**.
- Forms a **zygote ($2n$)** with a full set of 46 chromosomes.
- Zygote undergoes **cleavage (mitotic divisions)** to form a **blastocyst**.



Embryo Development and Migration

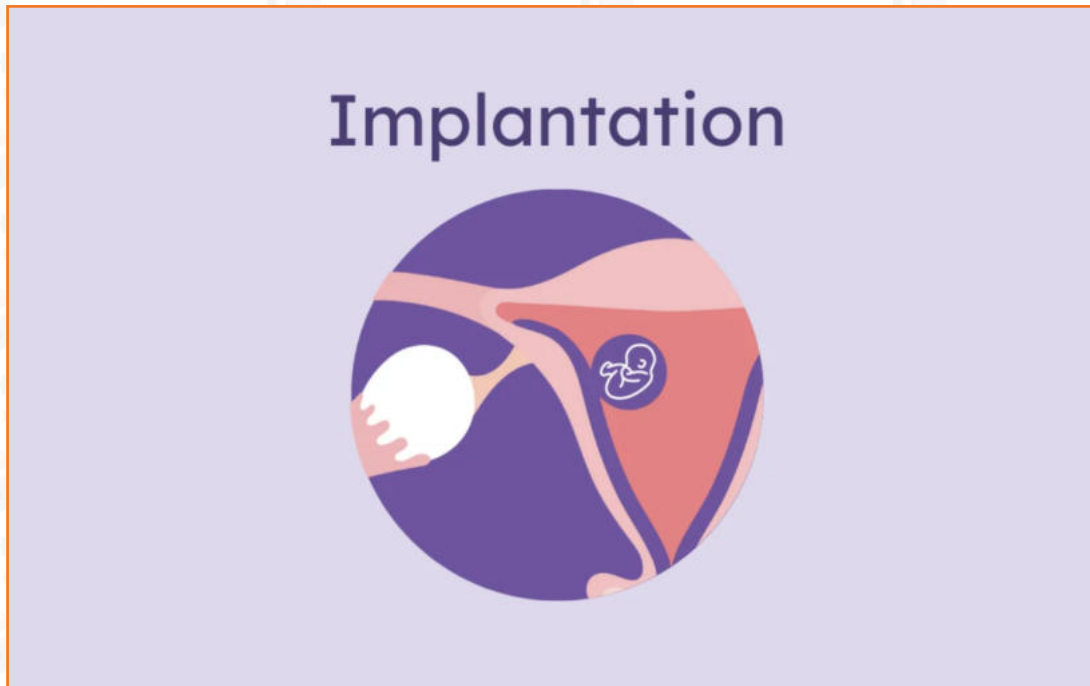
- The developing embryo moves through the **oviduct** toward the **uterus**.
- Takes about **5-7 days** to reach the uterus.
- Once in the uterus, the **blastocyst implants into the endometrium**, marking the start of pregnancy.





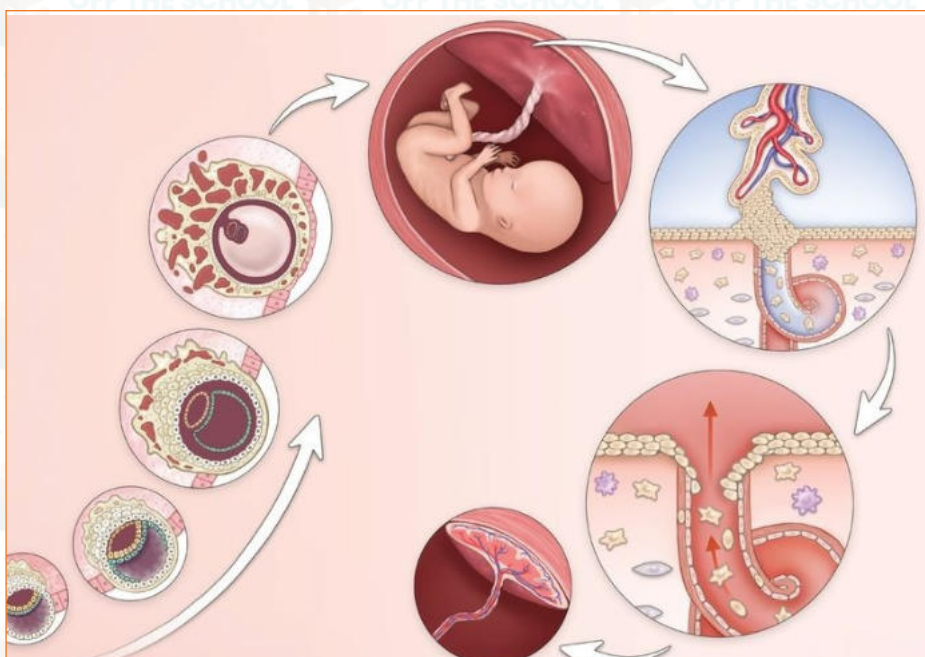
Implantation

- Implantation occurs in the **endometrial lining** of the uterus.
- The endometrium has been thickened and vascularized under **progesterone influence** (from the corpus luteum).



Placenta Formation

- **Placenta** forms from **both maternal and embryonic tissues**.
- It is a **discoid organ** that:
 - Facilitates **gas exchange**.
 - Transfers **nutrients** and **wastes** between maternal and fetal bloodstreams.
 - Acts as an **endocrine organ**.





Hormonal Regulation During Pregnancy

Human Chorionic Gonadotropin (hCG)

- Secreted by **early embryo/placenta**.
- Maintains the **corpus luteum** in the ovary.
- **Corpus luteum continues producing progesterone and estrogen** during early pregnancy.

Progesterone & Estrogen

- Initially produced by corpus luteum; later taken over by the **placenta**.
- Functions:
 - Maintain **endometrial lining**.
 - Inhibit **FSH and LH** from the anterior pituitary to prevent **ovulation of new eggs** during pregnancy.

STAGE	EVENT
Fertilization	Sperm + Egg → Zygote
Cleavage	Zygote → Multiple cells → Blastocyst
Implantation	Blastocyst embeds in uterine lining
Placenta Formation	Maternal + embryonic tissues → Exchange + hormone production
Hormonal Control	hCG maintains corpus luteum; later placenta produces progesterone & estrogen

Fertilization

- Occurs in the **oviduct (fallopian tube)**.
- A **sperm cell fuses with the secondary oocyte**, forming a **zygote (2n)**.
- The zygote undergoes **cleavage (rapid mitotic divisions)** to form a **blastocyst**.

Implantation

- **Blastocyst** travels to the **uterus** within 5–7 days after fertilization.
- **Implantation**: The blastocyst embeds itself in the **endometrium**, which has thickened due to **progesterone**.
- Successful implantation marks the beginning of **pregnancy**.



Estrogen

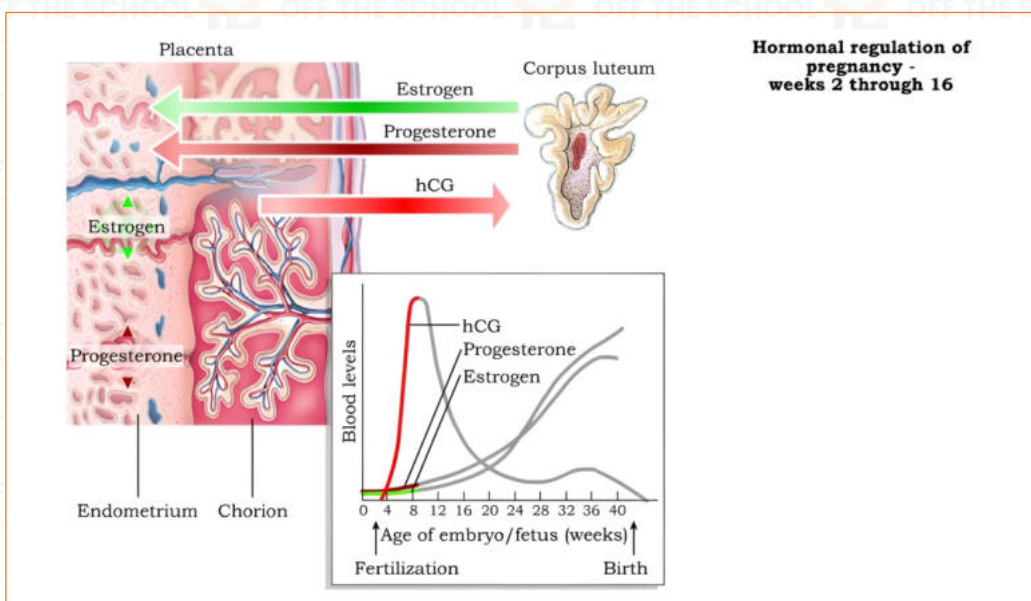
- Produced by **ovarian follicles** and later the **placenta**.
- Functions:**
 - Development of **female reproductive organs**.
 - Development of **secondary sexual characteristics**:
 - Breast development.
 - Wider pelvic girdle.
 - Distribution of **subcutaneous fat**, contributing to a **curvier body shape**.
 - Body hair pattern in females.
 - Stimulates the **growth of the endometrial lining** during the **proliferative phase** of the menstrual cycle.

Progesterone

- Secreted by the **corpus luteum** and later by the **placenta**.
- Functions:**
 - Maintains the **endometrial lining** for **implantation** and **pregnancy**.
 - Supports **breast development**.
 - Inhibits contraction of the **uterine muscles** to prevent early labor.
 - Regulates the **secretory phase** of the uterine cycle.

Hormonal Control of the Female Reproductive System

- Regulated by the **hypothalamus-pituitary-ovary axis**:
 - GnRH** (hypothalamus) → stimulates **FSH & LH** (anterior pituitary).
 - FSH** → follicle growth & estrogen secretion.
 - LH** → ovulation & progesterone secretion via corpus luteum.
 - Estrogen & progesterone** → negative feedback to inhibit FSH/LH (except just before ovulation).





Disorder of the Reproductive System

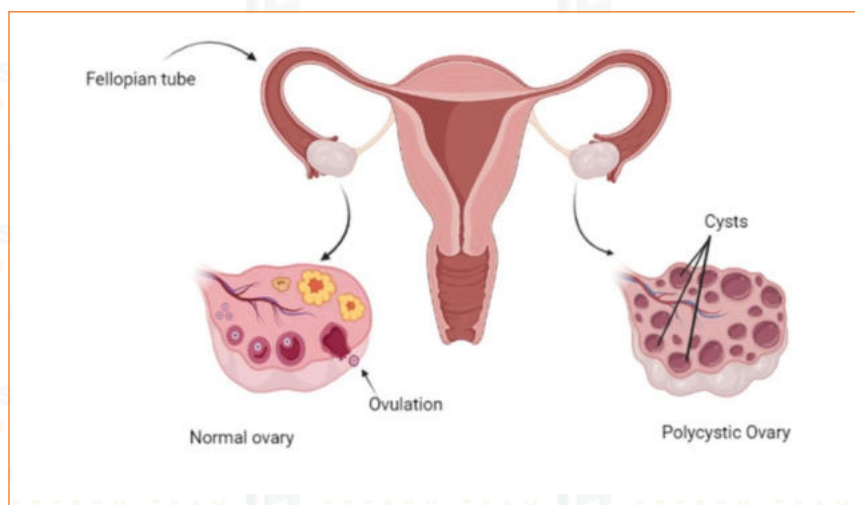
The **human reproductive system** can be affected by various disorders, which may impact fertility, hormonal balance, and sexual function. These conditions can result from **genetic, environmental, or lifestyle factors**.

Infertility

- **Definition:** Inability to conceive a child after **one year** of regular, unprotected sexual intercourse.
- **Can affect:** Both **males and females**.

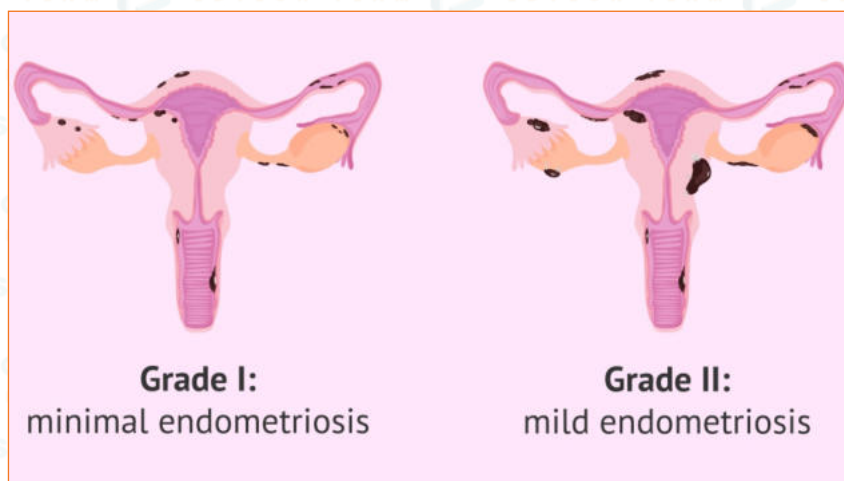
1. Polycystic Ovary Syndrome (PCOS)

- Hormonal imbalance causing enlarged ovaries with cysts.
- Symptoms: Irregular periods, acne, obesity, infertility.



2. Endometriosis

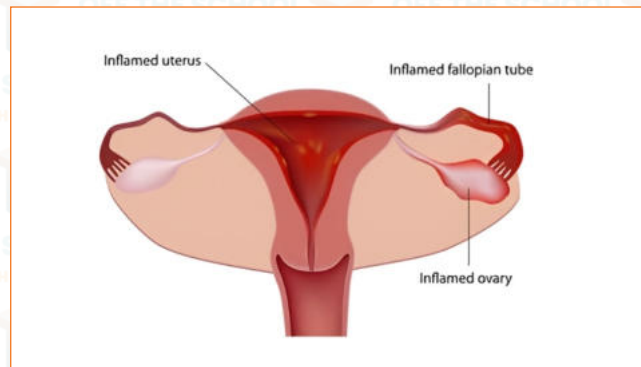
- Endometrial tissue grows outside the uterus.
- Symptoms: Pelvic pain, heavy bleeding, infertility.





3. Pelvic Inflammatory Disease (PID)

- Bacterial infection of female reproductive organs.
- Symptoms: Pain, fever, abnormal discharge; may lead to infertility.



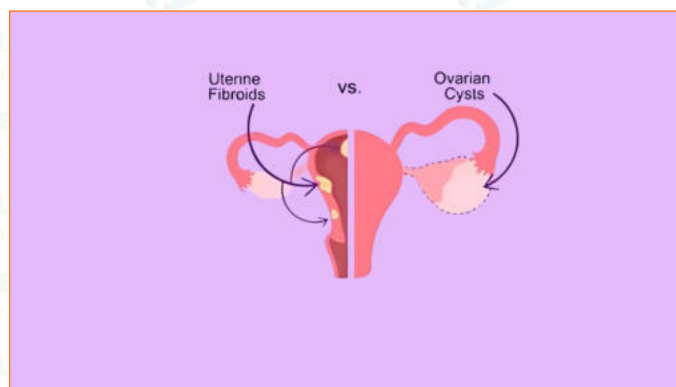
4. Premenstrual Syndrome (PMS)

- Physical and emotional symptoms before menstruation.
- Symptoms: Mood swings, bloating, fatigue.



5. Ovarian and Uterine Cysts or Fibroids

- Benign growths in ovaries or uterus.
- May cause pain, heavy bleeding, or infertility.



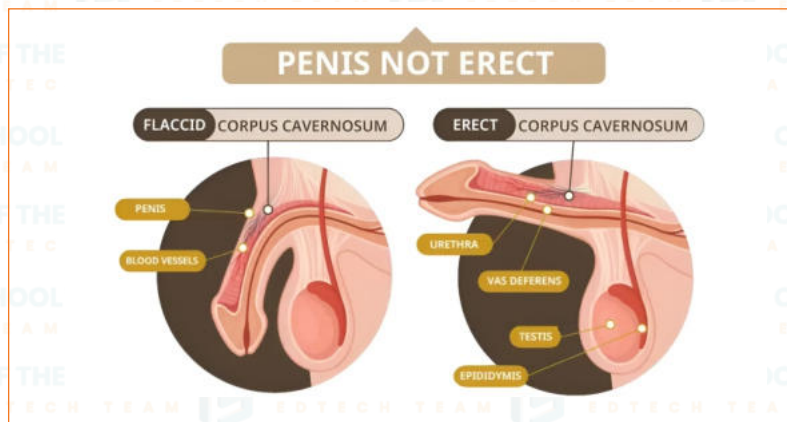


Common male reproductive disorders

1. Erectile Dysfunction (Impotence)

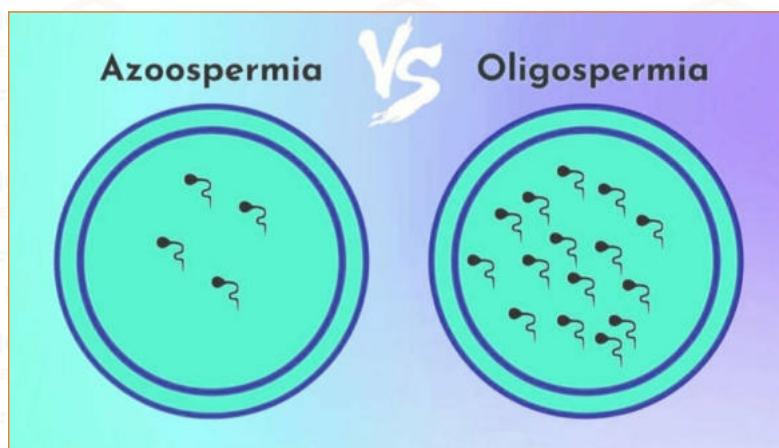
Inability to achieve or maintain an erection.

Causes: Psychological stress, poor circulation, hormonal imbalance.



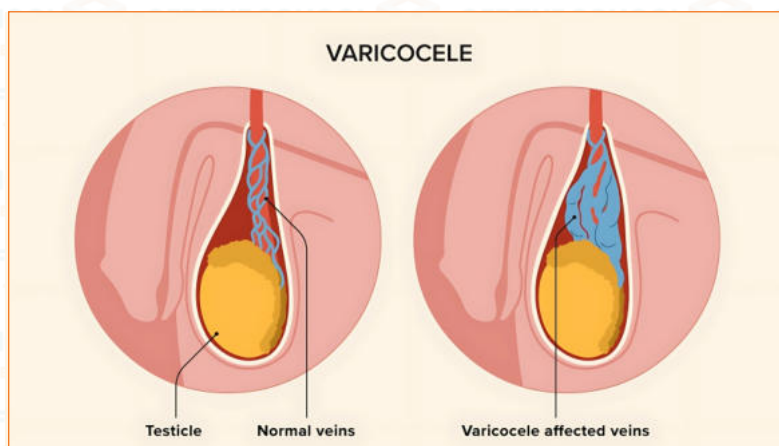
2. Oligospermia / Azoospermia

- Low sperm count / absence of sperm.
- Causes: Hormonal issues, infections, varicocele.



3. Varicocele

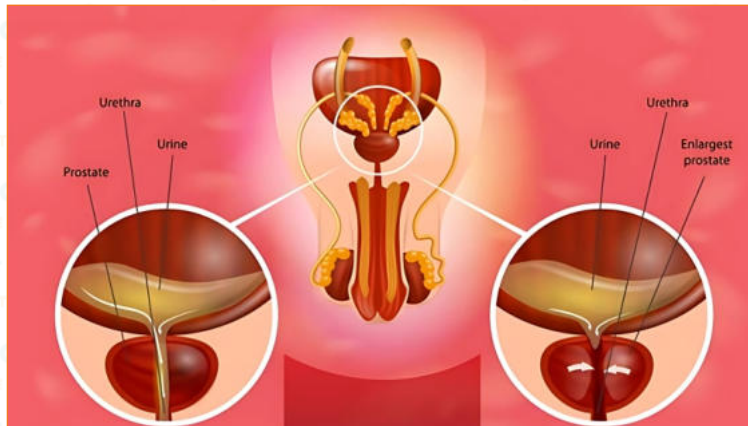
- Enlarged veins in the scrotum.
- May lead to infertility due to reduced sperm quality.





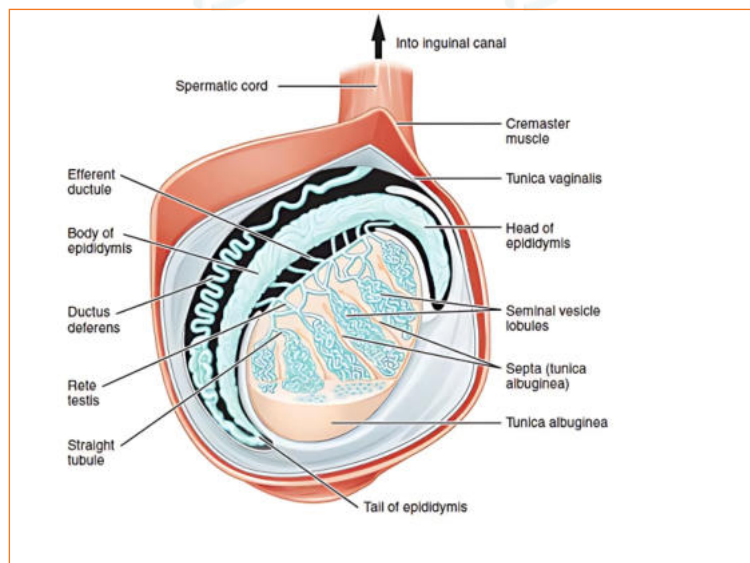
4. Prostatitis

- Inflammation of the prostate gland.
- Symptoms: Painful urination, pelvic discomfort.



5. Testicular Disorders

- Includes infections (orchitis), torsion, or cancer.
- May affect fertility and hormonal function.



Diagnosis Methods

- Blood hormone levels (FSH, LH, estrogen, testosterone)
- Ultrasound / imaging tests
- Semen analysis (for males)
- Laparoscopy / hysteroscopy (for females)

Treatment Options

- **Medications:** Hormone therapy, antibiotics (for infections), ovulation inducers.
- **Surgery:** To remove cysts, fibroids, or correct structural issues.





Causes of Male Infertility

a) Abnormal Sperm Production or Function:

- **Semen Analysis:** Microscopic examination assesses sperm quantity, concentration, motility, and morphology.
 - **Sperm Count:** The total amount of sperm in the ejaculate. A count under 20 million is typically considered low.
 - **Oligospermia:** Low sperm count.
 - **Azoospermia:** Complete lack of spermatozoa in the ejaculate, which can cause infertility.

b) Ejaculation Disorders:

- **Retrograde Ejaculation:** Semen enters the bladder instead of being expelled.
- **Erectile Dysfunction:** Difficulty achieving or maintaining an erection can also result in infertility.

c) Obstruction:

- **Blockages in the Reproductive Tract:** Can occur due to:
 - **Congenital Absence of the Vas Deferens:** Birth defect where the vas deferens is absent or blocked.
 - **Scarring:** Can be caused by previous infections or surgeries, preventing sperm transport.

d) Lifestyle Factors:

- **Excessive Alcohol Consumption:** Can affect sperm production and function.
- **Smoking and Drug Use:** Both can reduce sperm quality and count.
- **Obesity:** Can disrupt hormone levels, impacting sperm production.
- **Exposure to Environmental Toxins:** Chemicals, pesticides, and pollutants can harm sperm.
- **Prolonged High Temperatures:** Frequent exposure to hot tubs, saunas, or hot environments can lower sperm production.

Causes of Female Infertility

a) Ovulation Disorders:

- **Irregular or Absent Ovulation:** Prevents the release of eggs necessary for fertilization.
- Causes include:
 - **Hormonal Imbalances**
 - **Polycystic Ovary Syndrome (PCOS)**
 - **Thyroid Disorders**
 - **Premature Ovarian Failure**





b) Fallopian Tube Blockage:

- **Blockages or Damage:** Can prevent the transport of eggs and sperm, making fertilization difficult.
- Common causes:
 - **Pelvic Inflammatory Disease (PID)**
 - **Endometriosis**
 - **Previous Pelvic Surgeries**

c) Uterine or Cervical Issues:

- **Abnormalities in the Uterus or Cervix:** Can affect sperm passage or implantation of a fertilized egg.
- Conditions such as:
 - **Uterine Fibroids**
 - **Polyps**
 - **Cervical Stenosis**

d) Endometriosis:

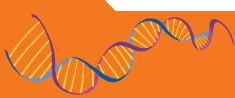
- **Tissue Growth Outside the Uterus:** The uterine lining grows outside the uterus, often affecting ovaries, fallopian tubes, and pelvic organs.
 - **Consequences:** Causes inflammation, scarring, and structural abnormalities, leading to infertility.

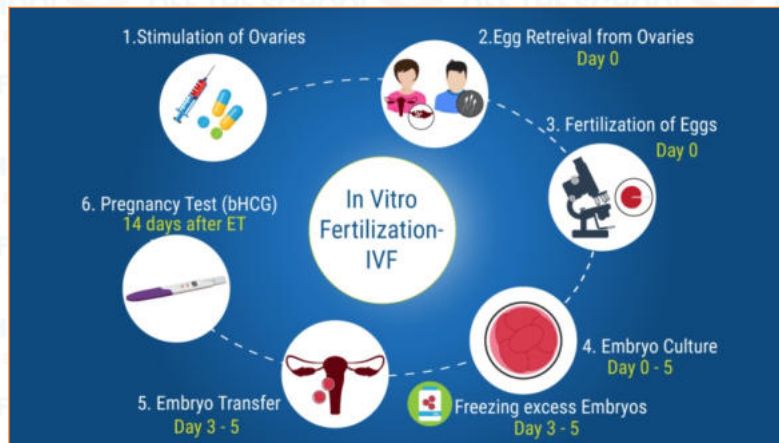
e) Age-related Factors:

- **Decline in Egg Quality and Quantity:** As women age, it becomes more difficult to conceive.
- Advanced maternal age is linked to:
 - **Increased Risk of Infertility**
 - **Higher Pregnancy Complications**

In Vitro Fertilization (IVF)

- **Definition:** IVF is a medical treatment where mature eggs are removed from a woman, fertilized with male sperm outside the body, and then implanted into the uterus.
- **Purpose:** Used for assisted reproduction in cases of infertility.
- **Illustration:** Fig. 20.13 depicts the process of IVF.





Miscarriage

- **Definition:** A miscarriage is the loss of a pregnancy before 20 weeks of gestation.
 - Occurs when the embryo stops developing, and the tissue is expelled from the body.

Causes of Miscarriage:

- **Placental Issues:** The embryo might implant near the cervix, causing the placenta to partially or completely cover the cervical opening, leading to bleeding and risk of miscarriage.
- **Haemorrhage:** Caused by the placenta detaching from the uterine wall.

Miscarriage vs. Abortion:

- **Miscarriage:** Unintended pregnancy loss due to natural causes (e.g., uterine or placental issues).
- **Abortion:** The intentional termination of a pregnancy, often due to health concerns or personal reasons.

Sexually Transmitted Diseases (STDs)

STDs are a group of diseases that can be transmitted through sexual contact with an infected person. Some of the key STDs are:





Gonorrhea

- **Cause:** Caused by the bacterium *Neisseria gonorrhoeae*.
- **Primary Affected Area:** Reproductive system (also affects the mouth, throat, and rectum).

Causes of Gonorrhea:

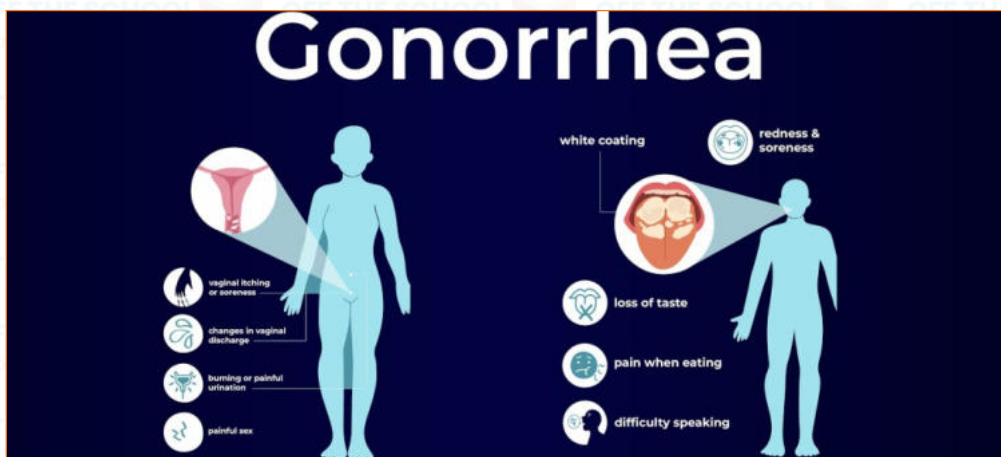
- **Transmission:** Primarily spread through sexual contact, including genital and oral contact with an infected person.

Symptoms of Gonorrhea:

- Painful or burning sensation during urination.
- Increased vaginal discharge in females and discharge from the penis in males.
- Pain or swelling in the testicles (males).
- Painful bowel movements or rectal itching.
- Sore throat or difficulty swallowing (oral gonorrhea).

Treatment of Gonorrhea:

- **Antibiotics:** Effective treatment with antibiotics. Due to growing antibiotic resistance, it's essential to complete the full course of prescribed antibiotics.



Syphilis

- **Cause:** Caused by the bacterium *Treponema pallidum*.
- **Primary Affected Area:** Various organs and systems, especially if untreated.

Causes of Syphilis:

- **Transmission:** Primarily spread through genital and oral sexual contact.

Symptoms of Syphilis:

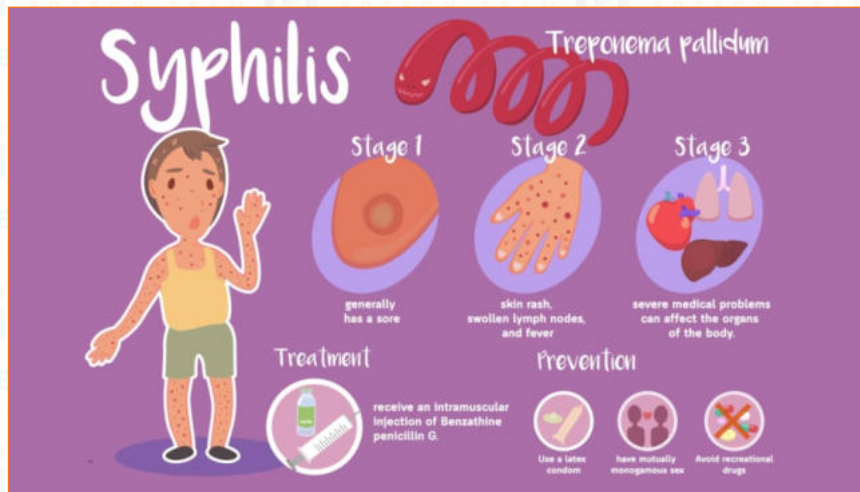
1. **Primary Stage:** Presence of a painless sore (chancre) at the infection site.
2. **Secondary Stage:** A rash and flu-like symptoms appear.
3. **Latent Stage:** Symptom-free period, but the infection is still transmissible.
4. **Tertiary Stage:** Severe complications affecting the heart, brain, bones, and other organs (neurological and cardiovascular problems).





Treatment of Syphilis:

- **Antibiotics:** Effective treatment with antibiotics, depending on the stage and severity.



AIDS (Acquired Immunodeficiency Syndrome)

- **Cause:** Caused by the **Human Immunodeficiency Virus (HIV)**, which attacks helper T cells (a type of lymphocyte).

Global Impact:

- **Epidemic:** AIDS has had a profound global impact, causing millions of infections and deaths.
- **Disproportionate Effect:** Particularly affects regions with limited healthcare, education, and resources, with Sub-Saharan Africa being the most impacted.
- **Global Concern:** Affects individuals of all ages, genders, and regions.

